

Each question is worth 5 points for a total of 120 points. The extra credit can get you an additional 5 points making the highest possible score 125 out of 120. Write answers CLEARLY. If I cannot determine your answer, the problem will receive a score of 0.

Problem 1

Convert each angle in degrees to radians. Express your answer as a multiple of π .

- a) 30°
- b) 90°
- c) 315°
- b) 125°

Problem 2

Let s denote the length of the arc of a circle of radius r subtended by the central angle θ . Find the missing quantity.

- a) $r = 17$ meters, $\theta = 3$, $s = ?$
- b) $r = 2$ inches, $\theta = ?$, $s = \pi$ inches
- c) $r = ?$, $\theta = \frac{\pi}{3}$, $s = 2\pi$ feet

Problem 3

Find the exact value.

- a) $\sin \frac{13\pi}{4}$
- b) $\cos \frac{7\pi}{6}$
- c) $\tan \frac{9\pi}{4}$

Problem 4

The point $(-5, 12)$ is on the terminal side of an angle θ . Find the exact value of the six trigonometric functions.

Problem 5

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $2 \cos^2 \theta + \cos \theta - 1 = 0$.

Problem 6

Let $\sin \theta = \frac{3}{4}$ where θ is in the first quadrant. Find the exact value of the other five trigonometric functions.

Problem 7

Find the exact value of $\tan(100^\circ)\cot(100^\circ)$.

Problem 8

Determine the amplitude and period of $y = -5\sin(\pi x)$.

Problem 9

Graph $y = 2\csc(2x - \frac{\pi}{2})$. Clearly label points on the graph used in constructing the graph.

Problem 10

Find the exact value of $\csc(\cos^{-1}\frac{3}{5})$.

Problem 11

Establish the identity and SHOW ALL YOUR WORK.

$$\frac{1 - \sin \theta}{1 + \sin \theta} = (\sec \theta - \tan \theta)^2.$$

Problem 12

Show that $\sec(\tan^{-1} v) = \sqrt{1 + v^2}$.

Problem 13

Find the exact value $\frac{\tan 20^\circ + \tan 25^\circ}{1 - \tan 20^\circ \tan 25^\circ}$.

Problem 14

Find the exact value $\sin[\sin^{-1} \frac{3}{5} - \cos^{-1}(-\frac{4}{5})]$.

Problem 15

Find the exact value $\tan \frac{9\pi}{8}$.

Problem 16

Establish the identity and SHOW ALL YOUR WORK.

$$\tan(3\theta) = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}.$$

Problem 17

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $2 \sin \theta + 3 = 2$.

Problem 18

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $2 \sin^2 \theta - \sin \theta - 1 = 0$.

Problem 19

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $\cos(2\theta) = \cos \theta$.

Problem 20

To measure the height of Lincoln's caricature on Mt. Rushmore, two sightings 400 feet from the base of the mountain are taken. If the angle of elevation to the bottom of Lincoln's face is 30° and the angle of elevation to the top is 45° , what is the height of Lincoln's face?

Problem 21

Solve the triangle $a = 6$, $b = 12$, $\alpha = 30^\circ$.

Problem 22

Find the area of the triangle $a = 4$, $b = 12$, $\gamma = \frac{\pi}{3}$.

Problem 23

Find the area of the triangle $a = 2$, $b = 2$, $c = 2$.

Problem 24

The displacement d (in meters) of an object at time t (in seconds) is given by the equation $d = 6 \cos(\pi t)$. Find the amplitude, period, and frequency.

Extra Credit

Solve for θ on the interval $0 \leq \theta < 2\pi$, where $\sqrt{3} \sin \theta + \cos \theta = 1$.